



Effects of Emergent Vegetation Patterns on Flow Velocity, Turbulence, and Erosion Around River Banks

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Abstract

The effects of two different emergent vegetation patterns on the flow velocity and turbulent kinetic energy (TKE) along the vegetated banks are experimentally investigated with natural reed stems. The results reveal that vegetation patterns can significantly alter the distributions of flow velocity and TKE. At locations for the maximum TKE, closed concentric circles named recirculation cells are observed. The location for the maximum TKE which may induce potential soil erosion for partly vegetated pattern moves forward 1 m in comparison with the vegetated pattern. An empirical function is found to describe the variations of TKE along the flow direction. Gentler distributed flow velocity and lower TKE at the rear part of the vegetation section for partly vegetated pattern suggest that vegetation pattern is an important factor for aquatic vegetation recovery in rivers. The findings of the research may provide some guidance for the engineering practices of river ecological restoration with aquatic vegetation.

Keywords Vegetation pattern · Flow velocity · Turbulence kinetic energy · Rigid unsubmerged vegetation · Soil erosion · Ecological river restoration

List of symbols

TKE	Turbulent kinetic energy
y	Longitudinal distance from the channel head
Re	Reynolds number
ρ	The water density at 15 °C
R	Hydraulic radius
μ	The dynamic viscosity of the water at 15 °C
Q	Flow discharge
B	Channel width
h	Water depth
Fr	Froude number
g	The gravitational acceleration
u	The transverse component of flow velocity
v	The longitudinal component of flow velocity
w	The vertical component of flow velocity

u'	The fluctuation of the transverse component of flow velocity
v'	The fluctuation of the longitudinal component of flow velocity
w'	The fluctuation of the vertical component of flow velocity
$\bar{u}$	The average value of the transverse component of flow velocity
$\bar{v}$	The average value of the longitudinal component of flow velocity
$\bar{w}$	The average value of the vertical component of flow velocity
TKE_0	The intercept values of the fitting curves for TKE
a, b	Constant coefficient of the fitting function for TKE
y_c	The longitudinal distance for the location where the TKE (or TKE_r) is highest
p	The significance level
R^2	The coefficient of determination
RMSE	The root-mean-square error
BIAS	The statistical bias
m	The number of measurements
TKE_j	The turbulent kinetic energy of the j th measurement
$\widehat{TKE_j}$	The predicted TKE of the j th measurement

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$\overline{\text{TKE}}$	The mean of TKE
TKE_r	Relative turbulent kinetic energy
TKE_{r0}	The intercept values of the fitting curves for TKE_r
a', b'	Constant coefficient of the fitting function for TKE_r

1 Introduction

As an integral part of aquatic ecosystem, vegetation provides many positive effects in habitat restoration, water pollutants purification, and bank soil erosion (Sand-Jensen 1998). In many countries, aquatic vegetation planting has become a popular technique in aquatic ecological restoration (Callow 2012). With this concern, numerous research works have been devoted to the comprehensive understanding of the role of vegetation in river ecosystems (Wu et al. 1999; Musleh and Cruise 2006).

In open-channel flow, flow velocity is an indicator measuring the transportation ratio of suspended particles, especially for non-scouring and non-silting velocities (Chow 1959; Abrari et al. 2015). As well recognized by researchers in the field, vegetation existence in river can alter the water surface profile and the distribution of flow velocity (Vanoni 2006; Neary et al. 2012). For the research focusing on the vegetation effects on flow structure in vegetated channel, many experimental studies have been conducted in laboratory flumes (Wu et al. 1999; Musleh and Cruise 2006). In recent years, to realize the mechanism of the interactive effects between aquatic vegetation and water flow, more attention has been paid to the research of the evolution of hydraulic roughness, biomechanics and streamlining of vegetation, and turbulent structure (Nepf 1999; Merritt and Cooper 2000; Zhang et al. 2015a). Bank vegetation increases the effective roughness height of the boundary, which has the effect of reducing the forces of drag and lift acting on the bank surface (Thorne 1990). Considering vegetation resistance is a function of vegetation size, structural properties, and location in the channel and local hydraulic conditions of water flow, Green pointed out that there is a nonlinear relationship between channel resistance and the proportion of the channel occupied by vegetation (Green 2005a). With the effects of vegetation resistance, aquatic plants can convert kinetic energy into TKE, which is an indicator measuring the development and recession of turbulence. Based on this relationship of energy transformation which should be responsible to wake generation and turbulence intensity, Nepf developed a model describing the drag, turbulence, and diffusion for flow through emergent vegetation (Nepf 1999).

In vegetated channels, the stems of rigid unsubmerged vegetation might cause local acceleration of flow velocity

and therefore turbulence moving in the wake zone (Thorne 1990), where detachment and entrainment usually occur (Leeder 1983). Turbulence can promote the exchange of momentum, energy, and matter; its effects are under high frequency and small scale (Clifford 1993). As well recognized, emergent vegetation in aquatic ecosystem can profoundly influence the turbulent structure and therefore alter the pattern of soil erosion and stream morphology (López and García 2001; Neary et al. 2012; Zhang et al. 2015b). The vegetation effects of different forms on turbulence characteristics are therefore becoming a research hot spot (Velasco et al. 2003; Afzalimehr and Dey 2009). Experimental tests were conducted by Velasco et al. (2003) using plastic plants seeded in a gravel bed to measure the turbulent regimes within and above flexible submerged vegetation. McBride et al. (2007) observed that TKE shows significant differences between forested and non-forested runs. Yagci et al. (2010) experimentally investigated the turbulence and kinetic energy patterns behind single erect vegetation, by which an empirical equation was proposed regarding the relationship between the kinetic energy components of flow. Neary et al. (2012) examined the interactions and feedback between flow, vegetation, and stream morphology with a variety of macrophyte patterns in riverine environments, and revealed that the spatial variability of the flow, momentum transfer, turbulence transportation are factors affecting the morphological processes in rivers.

In natural rivers, vegetation cover increases the irregularities of water flow and makes the flow systems partially continuous (Zhang and Dai 2009). The interactive effects among the water flow, aquatic vegetation type, distribution pattern, and soil erosion increase the difficulty of the research in the field (McBride et al. 2007; Baghlani and Talebbeydokhti 2013; Johari and Nejad 2015). Particularly, computations of experimental values of TKE are difficult to obtain due to a large number of measuring locations that need to be obtained (López and García 2001). Above all, little regard is given to the spatial affecting process of vegetation on flow velocity, turbulence, and erosion, especially for the local effects in open channel with different vegetation patterns.

This research intends to investigate how emergent vegetation patterns alter the transportation of flow velocity and TKE around the vegetated channel banks. With series experiments conducted in a laboratory channel which is, respectively, planted with vegetated and partly vegetated patterns, the distributions of flow velocity and TKE along the vegetated banks will be investigated. The research will also try to establish an empirical approach for predicting the extent of the TKE alternations. The primary goals of the research are to explore the mechanism and empirical relationships of emergent vegetation patterns on TKE in vegetated open channel and provide some guidance to the soil erosion control in ecological restoration practices for river banks.



2 Laboratory Experiments and Analytical Methodology

2.1 Experimental Setup

The primary goal of the research is to study the flow velocity and turbulent kinetic energy and further analyze the erosion possibility near the vegetated wall of the experimental channel under vegetated and partly vegetated patterns. From this point, a series of experiments are conducted in a laboratory channel whose bed and walls are all made of concrete. The internal dimensions of the experimental channel are 30 m in length, 1.3 m in width, and 0.8 m in depth, and the bed slope is fixed at 0.0055 m/m. In this experimental channel, water flow is circularly supplied by a piping centrifugal pump (Model: 200-200(I) A) which has a maximum flow discharge of 99.4 l/s. In this research, the flow discharge is select as 90 l/s to fully measure the internal flow and turbulent structure. In order to stabilize the water flow and control the flow discharge, a HQC-2 water level-discharge controller, a thin-walled rectangle weir, a stilling basin, and an energy dissipation fence are equipped in front of the main channel. As shown in Fig. 1, the 6-m-long test section is located between the $y=4$ m and 10 m locations away from the channel head, and there is a 4-m-long vegetation section within this test section.

Natural reed stems are used to simulate the rigid unsubmerged vegetation to make the results of the research

closer to reality. The experimental stems are cut from the bottom of natural reed stalks with a uniform length of 0.6 m and a mean diameter of 0.007 m. For the vegetation planting, a 0.2-m-deep pool is built concave downward the channel bottom in the vegetation section ($y=5-9$ m). Then, the pool is filled with fine sand and covered with punched plates on which several holes with a uniform diameter of 0.01 m are drilled in a uniformly stagger array. As such, stems can be uniformly staggered and firmly planted into these holes according to the experimental requirements. To get the experimental conditions of vegetated and partly vegetated patterns, the experimental stems are planted at the bottom of the channel by two arrays. As shown in Fig. 2a, the vegetated pattern can be formed by planting reed stems in the whole 4-m-long section ($y=5-9$ m). The area within the red pane in Fig. 1 will not be planted which means the reed stems are only planted in half of the 4-m-long section ($y=5-9$ m), and a partly vegetated pattern can be obtained (see Fig. 2b). In this research, the vegetated and partly vegetated experimental conditions are, respectively, defined as P1 and P2. As the flow velocity can be highly reduced by dense vegetation, higher vegetation densities are generally used in the practices of river ecological restoration (Nikora et al. 2013). In this research, the reed stems are planted with a density of 202 stems/m² for both of these vegetation patterns.

With the aim of the research which is to realize the velocity and turbulence variations of the water flow affected by emergent vegetation patterns, 3D flow velocities (u , v , and

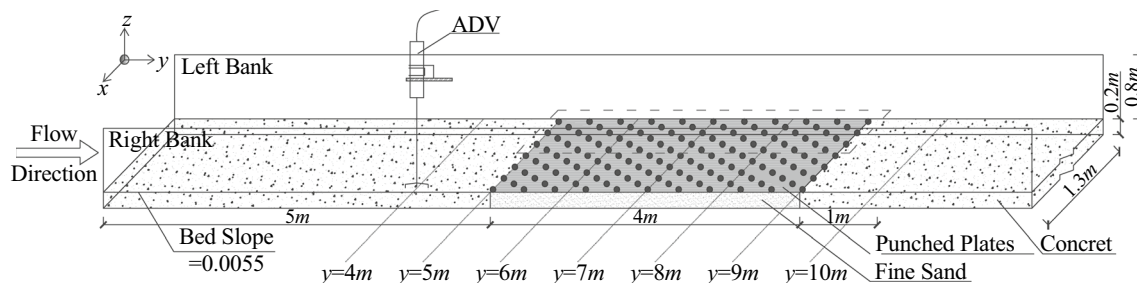
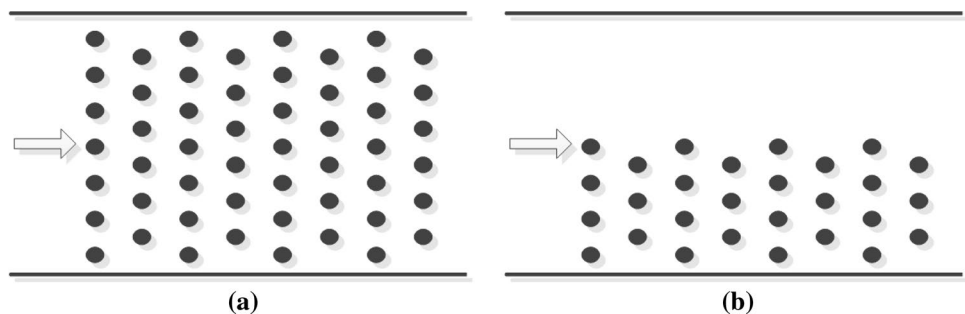


Fig. 1 Experimental channel setup

Fig. 2 Vegetation patterns: **a** vegetated pattern-P1, **b** partly vegetated pattern-P2



w corresponding to x , y , z directions along the channel width, length, and depth which can be seen in Fig. 1) are measured by 16-MHz 3-Dimensional Micro Acoustic Doppler Velocimeter (ADV, Provided by Son Tek YSI Inc., San Diego, California, see Fig. 3) whose resolution is 0.1 mm/s. As a non-contact velocimeter, ADV can provide enormous convenience for analysis of flow and turbulent characteristics due to its very small disturbance to the water flow and high measuring accuracy (1% of the measured range). In the present research, the maximum measurement error for flow velocity should be lower than 0.01 m/s as the measurement range for ADV is selected at 1.00 m/s.

For gaining reliable and stable data with high efficiency, the sampling frequency and testing duration for flow velocity are, respectively, set at 50 Hz and 60 s, which are sufficient enough to obtain a stable time-averaged velocity for both conditions of channel with and without reeds stems (Jing et al. 2010). A total of 3000 samples can therefore be collected at each measure point for the flow velocities, in the experiments performed. It should be noticed that each ADV has three acoustic receivers and one acoustic transmitter, providing water velocity measurements in three directions: transverse component u , longitudinal component v and vertical component w . In addition, the signal-to-noise ratio of each flow velocity measurement should be maintained above 15 to ensure the quality of the experimental data collected.

For two different vegetation patterns, the flow velocities are measured along the vertical profile 0.05 m away from the vegetated banks (right bank from the view of flow direction, which is the bank on the vegetated side for P2) to investigate the turbulence structure around the channel banks (see Fig. 1). All the measuring points are 0.05 m away from the vegetated walls, which are the positions between the wall and the reed stems nearest the wall. This position is not too near to the wall in order to avoid that the effects of the roughened wall are much higher than the vegetation patterns, and also not too far from the wall to ensure that the measurements can be used to analyze the erosive possibility due to the interactions between vegetation patterns and water flow. As can be seen in

Fig. 1, there are seven measure sections within the 6-m-long test section. And along the vertical direction, the flow velocities are measured every 0.005 m from the channel bed to the water surface for each measure location.

The water depths along the vegetated banks of two different vegetation patterns are also measured by ABF2-2 Water Level Measuring System (Provided by School of Water Resources and Hydropower Engineering at Wuhan University, Wuhan, Hubei, China). ABF2-2 can measure the water level point by point with a uniform interval, automatically. The measuring resolution and accuracy are 0.01 mm and 0.04 mm, which matches the requirement of the experiments. In this research, the measurements interval along the flow direction for water depth is 0.1 m. In order to avoid the uncertainty of the measurement, the water depth is measured 3 times at each measuring point. And the water depth h used in the analyses of the research can therefore be obtained by averaging these three measurements.

2.2 Determinations of Hydraulic Parameters

Reynolds Number and Froude Number along the vegetated banks for vegetated and partly vegetated patterns are first calculated to determine the flow conditions in the channel. The Reynolds Number can be obtained at every measure point all over the channel, the calculating formula is

$$Re = \rho v R / \mu = \rho Q / (\mu (B + 2h)) \quad (1)$$

where $\rho = 0.9991 \times 10^3 \text{ kg/m}^3$ is the water density at 15 °C, v is the streamwise flow velocity along the y direction, $\mu = 1.1404 \times 10^{-3} \text{ Pa s}$ is the dynamic viscosity of the water at 15 °C, $Q = 90 \text{ l/s}$ is the flow discharge, $B = 1.3 \text{ m}$ is the width of the experimental channel. The hydraulic radius R and the water depth h in Eq. (1) can be, respectively, calculated with

$$R = Bh / (B + 2h) \quad (2)$$

and

$$h = Q / vB \quad (3)$$

Fig. 3 Actual views of the experimental channel and the measuring instrument of ADV: **a** without vegetation, **b** with vegetation



In this research, the Reynolds Numbers for all the measuring points belong to the ranges of 43, 555–47, 136, 46, 806–49, 142 for the vegetation patterns of P1 and P2, respectively. All Reynolds numbers are much higher than the critical value 500 for open-channel flow, which illustrates that the flow conditions for the experiments are fully developed turbulent.

The formula used for the calculation of the average Froude number is

$$Fr = v/(gh)^{0.5} \quad (4)$$

where $g = 9.81 \text{ m/s}^2$ is the gravitational acceleration. And the Froude Numbers for the vegetated and partly vegetated experiments are 0.21 and 0.32, respectively. For both of the two experimental conditions, the Froude numbers are below 1.0, which attests that the flows in the channel are the state of the gradually varying flow.

As mentioned above, the 3-D flow velocities along the banks of vegetated and partly vegetated patterns are measured in the experiments, with which the total flow velocities can be determined with

$$V = (u^2 + v^2 + w^2)^{0.5} \quad (5)$$

where u , v and w are the transverse (along the channel width), longitudinal (along the channel length) and vertical (along the channel height) component of flow velocity, respectively.

As the mean kinetic energy per unit mass associated with eddies in turbulent flow, the turbulence kinetic energy (TKE) is calculated and analyzed to investigate the influence of vegetation patterns on turbulent characteristics. With the instantaneous flow velocities measured in the experiments, the TKE can be calculated at each measure point for both vegetated and partly vegetated cases. In turbulent flow, TKE is an isotropic parameter characterized by measured root-mean-square velocity fluctuations. For each measurement point, TKE can be quantified by the mean of the turbulence normal stresses as below:

$$\text{TKE} = (\overline{u'^2} + \overline{v'^2} + \overline{w'^2})/2 \quad (6)$$

where $u' = u - \bar{u}$, $v' = v - \bar{v}$, and $w' = w - \bar{w}$ represent the fluctuations of the transverse, longitudinal, and vertical component of flow velocity u , v , w , which means the deviations of instantaneous measurements are different from the average, respectively (Hiscoxa et al. 2015; Zhang et al. 2015b). $\bar{u}$, $\bar{v}$, $\bar{w}$ are the average values of u , v , w , and the average processes above can be made by averaging the values of 3000 samples measured by ADV, accordingly. In addition, the relative turbulent kinetic energy which is defined as $\text{TKE}_r = \text{TKE}/\text{TKE}_{\max}$ will also be analyzed regarding the longitudinal locations for different vegetation patterns.

3 Results and Discussion

3.1 Bank of Vegetated Pattern (P1)

3.1.1 Distributions of Water Depth and Flow Velocity

Before investigating the variations of TKE along the channel banks, the distributions of flow velocity and water depth are first analyzed in this research. In Fig. 4, the longitudinal distributions of relative flow velocity and water depth along the bank of vegetated pattern are plotted. It should be noticed that the variation trends for flow velocity and water depth are not completely negatively correlated as the measurement points are near the channel bank.

As can be seen in the figure, for the vegetated pattern, vegetation block can raise the water depth and reduce the flow velocity before the vegetation section. In natural rivers, reduction in flow velocity and therefore erosion power will reduce the erosive effects on river banks and may induce suspended particle deposition although the non-scouring velocities are different depending on the boundary materials (Chow 1959). These effects are beneficial to the control of bank soil erosion, which have promoted the aquatic vegetation planting and management to be a common technique for river ecological restoration, especially for the soil erosion control (Zhou et al. 2006). However, deposition of suspended particles will occur, and the entrance of vegetation section may also be silted up if the flow velocity is too small, which is similar to the results reported in previous studies (Nagle 2002; Butt et al. 2010). Planting the vegetation stems with various densities may be a possible way to uniformize the flow velocity along the flow direction.

In the vegetation section, due to the raising of water level at the entrance of vegetation section, flow velocity along the channel bank in this case increases all along and reaches a maximum value at the exit of the vegetation section. Although the flow velocity presents a decreased tendency after the vegetation section, the values of flow velocity

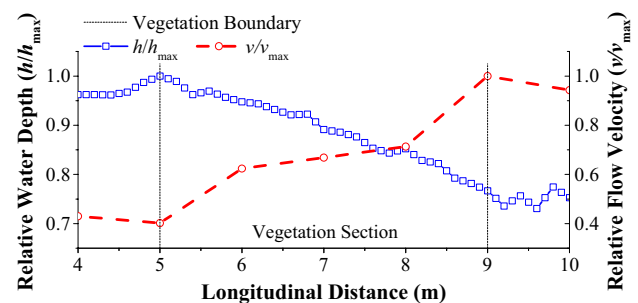


Fig. 4 Variations of the relative flow velocity and water depth along the bank of P1

around the exit of vegetation section are higher. This conclusion is in good agreement with previous studies (Green 2005b; Zhang et al. 2015b). As reported in their studies, the flow velocity should be extremely high at the exit of vegetation section. As per the concept of non-scouring velocity, the area before and behind the exit of vegetation section should be noticed for the bank soil erosion. In addition, a comprehensive analysis on TKE along the bank of vegetated channel will be performed, because that turbulence distribution has significant effects on the suspended particles and soil erosion (Huppert et al. 1993).

3.1.2 Vertical Profile of TKE

The vertical profiles of TKE along the vegetated banks are selected to characterize the turbulent structure around river banks with different vegetation patterns. For the bank along the vegetated pattern, the contours of TKE are drawn in Fig. 5 using the inverse distance weighting method. As can be found in the figure, the variation of TKE along the flow direction is significantly affected by vegetation block and can be divided into largely three sections. In addition, the interfaces among these three sections are clear and almost upright.

Firstly, the measured TKE values are small and stable in the section $y=4$ m (1 m before the vegetation block) to $y=7.2$ m (near the middle of the vegetation block), which forms a low-TKE zone. Within this zone, the TKE values are relatively higher at the area near the bottom of $y=5$ m location, where the TKE values increase from 53.0 to 154.9 m^2/s^2 and form several closed circles reflecting the complex interaction between the vegetation and water flow. This phenomenon may be caused by the higher momentum exchange at the entrance of vegetation section where the water flow breaks against the vegetation block. Outside this region, the TKE in the section $y=4$ – 7.2 m is mostly lower than 53.0 m^2/s^2 , indicating that the turbulence variation is very small. Considering that the flow velocity will be reduced at the forepart of the vegetation section due to the vegetal

blocking effects (Green 2005b), the potential risk of soil erosion should be lower in this section. The probability of potential soil erosion should therefore be very small (Toorman et al. 2002).

Secondly, in the section of $y=7$ – 9.9 m, the TKE values increase rapidly along the flow direction from 53.0 m^2/s^2 and reach the maximum value 562.5 m^2/s^2 at the location of $y=9$ m (the exit of the vegetation section). After $y=9$ m, the TKE values begin to decrease until 78.4 m^2/s^2 at $y=9.9$ m location. At $y=9$ m location, there are four regions along the water depth where TKE are much higher and form several closed concentric circles. Among these regions, the largest one is near the water surface although the water depth at this location should be very small, and the highest value of TKE is 562.5 m^2/s^2 . At the centers for the other three areas, the TKE values are around 200 m^2/s^2 . This phenomenon is in good agreement with the results reported by Curran (Curran and Tan 2014). In their research, such circles were named as recirculation cells, and the calculated TKE revealed that the clusters increased turbulence and induced the formation of recirculation cells. Similar to the $y=5$ m location, $y=9$ m location is also a boundary of the vegetation block, while the difference is that the TKE values are much higher than that of 5 m location. As reported in previous studies, the flow velocity varies rapidly at the interfaces of vegetation block in open channel and therefore momentum exchange, especially for the end part of vegetation section (Nikora et al. 2013). Much higher TKE in this section demonstrates higher momentum exchange and turbulence variation, the erosive effects on the bank can therefore be amplified due to the higher turbulence intensity (McBride et al. 2007). Along with the decrease of the vegetal blocking effects, the probability soil erosion should be higher as the flow velocity increases in this section.

At last, after $y=9.9$ m location, the TKE values decrease toward the level of low-value zone.

The results above illustrate that the TKE transportation is much higher at the exit of the vegetation section, and may promote the potential soil erosion risk at this area.

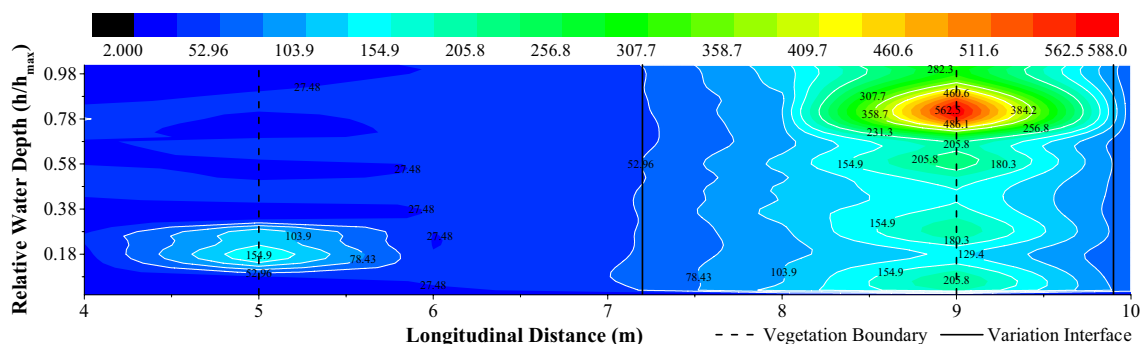


Fig. 5 Vertical profile of TKE along the bank of P1

This situation is similar to the effects of hydraulic structural work, which will also induce a serious local influence on accelerating channel erosion at the back area (Vanoni 2006). This may suggest that the channel bank around the rear part of vegetation section needs to be enhanced in the practice of river ecological restoration with aquatic vegetation.

3.2 Vegetated Bank of Partly Vegetated Pattern (P2)

3.2.1 Distributions of Water Depth and Flow Velocity

Figure 6 shows the variations of relative flow velocity and water depth along the vegetated bank of partly vegetated case in this research. For the vegetated bank of P2, the variation tendencies of flow velocity and water depth are similar to that of vegetated channel. As can be seen in the figure, the variations of flow velocity also present the tendencies as follow: decrease before vegetation block, increase all along the vegetation section, and decrease after vegetation block. The variation trend of water depth is opposite to that of flow velocity, although the variations of flow velocity and water depth are not completely negatively correlated. What can be concluded similar to the results for the vegetated pattern is that the reduction in flow velocity at the fore part of

vegetation section may reduce the erosive effects on river banks, which are beneficial to the control of bank soil erosion. At the region near the exit of the vegetation section, the flow velocity is much higher which may induce higher soil erosion.

3.2.2 Vertical Profile of TKE

The vertical profile of the TKE along the vegetated bank of partly vegetated channel is shown in Fig. 7, which is also constructed with the inverse distance weighting method. Different from the bank of vegetated pattern, the variation of TKE along the vegetated bank of partly vegetated channel becomes more complex. It can be divided into four sections on the whole, although the interfaces among these sections are not very clear.

For the part before the vegetation section ($y=4\text{--}5\text{ m}$), the TKE is lower than $44.9\text{ m}^2/\text{s}^2$, while the values for the section where water just flows into the vegetation block ($y=5\text{ m}$ –around 6.5 m) are even lower than $24.4\text{ m}^2/\text{s}^2$. Similarly, in considering the lower flow velocity at the forepart of vegetation section, the potential soil erosion risk should be relatively small in this section.

Since $y=5\text{ m}$ location, the TKE increase gradually and reach the maximum values $228.8\text{ m}^2/\text{s}^2$ at the location of $y=8\text{ m}$ location. As can be seen in the contour map, the vertical distribution of TKE at the $y=8\text{ m}$ location is similar to the $y=9\text{ m}$ location of the vegetated bank of P1. There are also several recirculation cells where TKE values are much higher and form a lot of concentric circles at each area. Differences are that these circles occupy a longer area in the channel and the maximum values for these areas are similarly close to $230\text{ m}^2/\text{s}^2$ which is lower than that of the vegetated bank of P1. From the perspective of locations, the location for the maximum TKE moves forward to the upstream at 8 m location, in comparison with the bank of the vegetated pattern, which can be explained by the flow velocity variations in the partly vegetated channel. As revealed in previous studies, for the partly vegetated reach, the flow

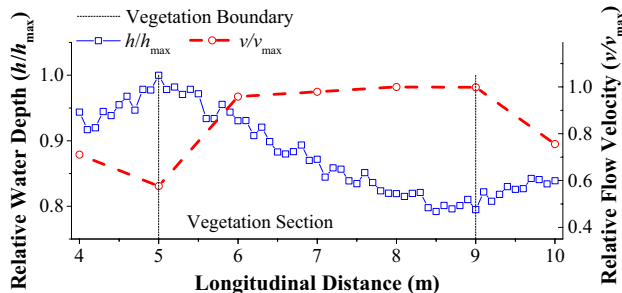


Fig. 6 Variations of the relative flow velocity and water depth along the vegetated bank of P2

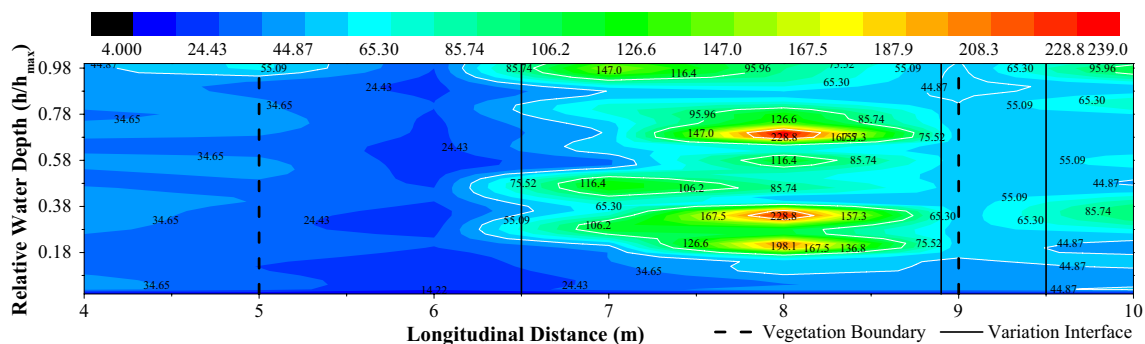


Fig. 7 Vertical profile of TKE along the vegetated bank of P2

velocity in the un-vegetated side is much higher than that of vegetated side. After this partly vegetated reach, the velocity difference between these two sides decreases rapidly (Zhang et al. 2015a). In other words, the water flow will be forced to the unvegetated side in the partly vegetated reach, and the area near the $y = 8$ m location for P2 can be considered as an exit of the vegetation block. The acutely turbulent area is therefore moved forward to the region near the $y = 8$ m location. After $y = 8$ m location, the TKE decreases rapidly until $75.5 \text{ m}^2/\text{s}^2$ at $y = 8.9$ m location. Same as that of the vegetated bank for P1, the TKE uses a 0.9-m-long section decreasing to relative low values. For the section $y = 5\text{--}8.9$ m, the higher TKE can also bring higher momentum exchange and turbulence transportation. Combining with the increasing flow velocity, this moved forward and expanded section should be the dangerous area for potential soil erosion, which should be taken care of in river vegetation recovery.

In the section $y = 8.9\text{--}9.5$ m, the TKE values are relatively low and distributed stably.

After $y = 9.5$ m location, an interesting phenomenon can be found is that the variation of TKE increases slightly. That can be explained by the velocity mixing when the vegetation array disappears. As mentioned above, the velocity difference between the vegetated and unvegetated side will decrease rapidly after the partly vegetated reach which makes the water flow mix dramatic. The mixing of flow velocity brings higher momentum exchange and therefore higher turbulence. This phenomenon suggests that longer area after the vegetation section should be noticed comparing with the whole vegetated pattern if the partly vegetated pattern is used for the river restoration.

3.3 Vegetation Patterns with Velocity Distribution

The distributions of flow velocity along the vegetated banks of vegetated and partly vegetated patterns are plotted in Fig. 8 to compare the effects of vegetation patterns. As shown in the figure, in the before part and fore part of vegetation section (before $y = 8$ m location), the flow velocities for vegetated pattern are lower compared with the partly vegetated pattern, and higher for the remaining part which is 1 m before and after the exit of vegetation block. Although the lower flow velocities in the forepart may be beneficial to the soil erosion control, the higher velocities around the exit of vegetation section are undesirable. In comparison with vegetated pattern, the variation of flow velocity for the partly vegetated channel is more stable, not too low at the entrance and not too high at the exit of vegetation section. Comprehensively, partly vegetation pattern may be more suitable for vegetation recovery in rivers, according to the present experimental conditions.

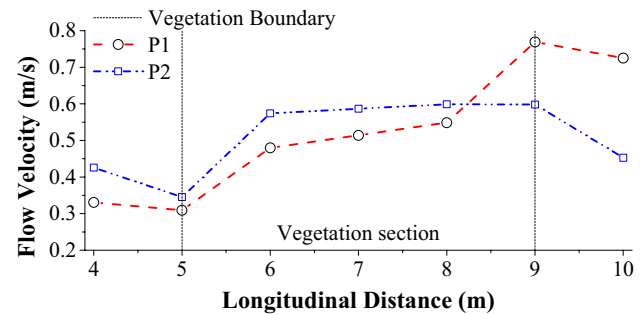


Fig. 8 Flow velocity distributions along the vegetated banks of P1 and P2

3.4 Vegetation Patterns with Average TKE

As recognized by researchers in this field, it is necessary to average the measurements taken in the vegetated channel on space scale to obtain meaningful observations (López and García 2001). In this research, the depth-averaged TKE for each measurement point is calculated and plotted in Fig. 9 to quantitatively investigate the variation of TKE along the vegetated banks affected by two different vegetation patterns.

As shown in Fig. 9, the curves of average TKE for two different vegetated banks seem unchanged or changed slightly before $y = 6$ m location. Same as the above results obtained for the vegetated banks of vegetated and partly vegetated patterns, the average TKE begin to increase after $y = 6$ m location and reach their maximum values at $y = 8$ m and $y = 9$ m locations, respectively. From the perspective of comparison, the values of average TKE for the bank of vegetated channel are mostly higher than that of the vegetated bank of partly vegetated channel. To some extent, this may indicate that the vegetation planting is not the more the better in the engineering practices of river ecological restoration. The planting pattern is also an important factor affecting the effects of the restoration.

Based on the results above, a further quantitative analysis is performed with OriginPro 9.0 in expecting to find a suitable method to describe and predict the variations of the average TKE along different banks. Firstly, it can be obviously observed from the figure that both the variations of the average TKE along the vegetated banks under vegetated and partly vegetated patterns present single peak forms. The fitting function can therefore be determined in the category of “Peak Functions” by the “Nonlinear Curve Fit” function in OriginPro 9.0. The detailed location for the “Peak Functions” can be found by the following operation steps: open OriginPro 9.0, click the menu “Analysis,” “Fitting,” “Nonlinear Curve Fit,” “Open Dialog” in turn. Secondly, the functional forms as well as the fitting results are compared between different functions to find the best one. Thirdly, the relevant parameters as well as the coefficients

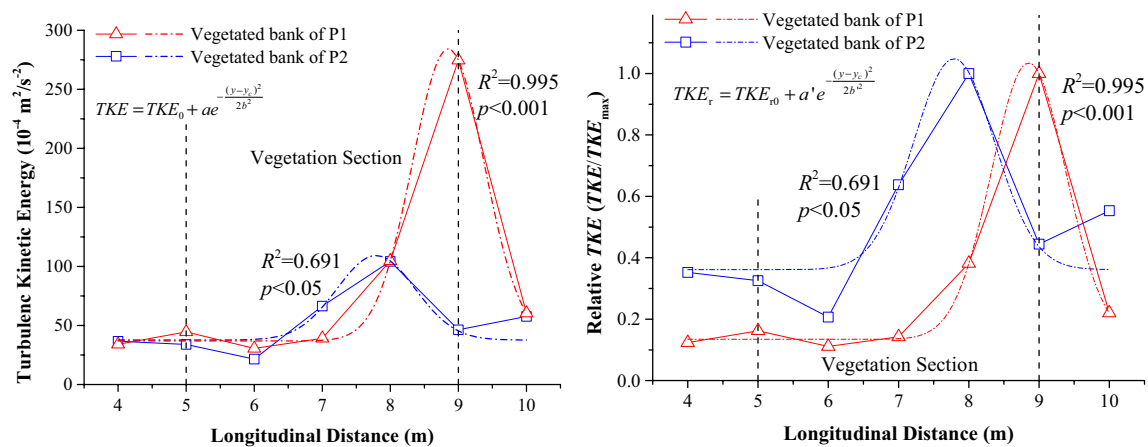


Fig. 9 Nonlinear fitting of average TKE and TKE_r along the vegetated banks of P1 and P2

of determination R^2 , p value, etc. can also be calculated together in the process of nonlinear curve fitting. The results demonstrate that the variations of TKE along the vegetated banks of vegetated and partly vegetated patterns can be expressed with Eq. (7) which is an amplitude version of Gaussian peak function. In this function, TKE_0 is the intercept value of each fitting curve, y_c represents the location of the peak where TKE is the highest, and the parameters a and b are constant coefficients.

$$TKE = TKE_0 + a * \exp(-(y - y_c)^2 / 2b^2) \quad (7)$$

In the present research, the parameters in the function are $TKE_0 = 36.94, 37.69$, $y_c = 8.85, 7.79$, $a = 247.21, 71.54$ and $b = 0.53, 0.57$ corresponding to the vegetated banks of vegetated and partly vegetated patterns. In addition, the coefficients of determination R^2 for the fitting lines corresponding to the vegetated banks of P1 and P2 are, respectively, 0.995 and 0.691, with the p -values of 1.00×10^{-4} and 1.27×10^{-2} , respectively. As both the p -values are less than the significance level of $p < 0.05$, the coefficient of determination R^2 , the root-mean-square error (RMSE) and BIAS can be utilized to characterize the nonlinear curve fitting effects (Najafzadeh and Saberi-Movahed 2018; Najafzadeh and Zeinolabedini 2018). R^2 , RMSE and BIAS can be determined by Eqs. (8), (9), and (10), respectively.

$$R^2 = 1 - \frac{\sum_{j=1}^m (TKE_j - \widehat{TKE}_j)^2}{\sum_{j=1}^m (\widehat{TKE}_j - \overline{TKE})^2} \quad (8)$$

$$RMSE = \sqrt{(1/m) * \sum_{j=1}^m (TKE_j - \widehat{TKE}_j)^2} \quad (9)$$

$$BIAS = (1/m) * \sum_{j=1}^m (\widehat{TKE}_j - TKE_j) \quad (10)$$

where m is the number of measurements, TKE_j and $\widehat{TKE}_j$ are the j th measured and predicted turbulent kinetic energies, respectively, and $\overline{TKE}$ is the mean TKE for all the measurements.

For the fittings with Eq. (7), the values of RMSE are 9.95 and 3.96 corresponding to the vegetated banks of P1 and P2, while the BIAS values are 1.00×10^{-4} and -4.90×10^{-4} , respectively. Based on the comprehensive considerations above, it can be concluded that these two fitting results show remarkable performance, which means that the fitting results demonstrate relatively high precisions to some extent. As can be found, the values of the parameter y_c where TKE is highest demonstrate agreement with the analysis above. For the coefficient b , the values are around 0.55 for the vegetated banks, this illustrates that the values for coefficient b may be an approximate value under a determined hydraulic condition. As such, Eq. (7) may be used to predict the TKE values along the vegetated banks for the similar vegetated and hydraulic conditions, if it can be verified and calibrated with more experimental results under different flow discharges, channel slopes, etc.

For a certain condition, a dimensionless equation should be more conducive. For the relative turbulent kinetic energy TKE_r , Eq. (11), which is a dimensionless form of Eq. (7), is utilized for the nonlinear fitting. That is

$$TKE_r = TKE_{r0} + a' * \exp(-(y - y_c)^2 / 2b'^2) \quad (11)$$

where $TKE_{r0} = 0.13, 0.36$, $y_c = 8.85, 7.79$, $a' = 0.90, 0.69$, and $b' = 0.53, 0.57$. It can be found that the values for y_c and b' are same as that of Eq. (7). In addition, the values of p and R^2 of these fitting lines are also the same as those of the fittings with Eq. (7).

The results of this research reveal that the vegetation pattern demonstrates important significance to the practices of river ecological restoration utilizing aquatic vegetation. In

addition to hydraulic conditions, it is necessary to consider the vegetation pattern together in the design of ecological engineering, in order to avoid the potential siltation at the entrance of vegetation section and the potential erosion at the exit. However, there are many other factors affecting the influence of vegetation on water flow in actual situations, further studies focusing on this subject are therefore necessary.

4 Conclusions

For the vegetated banks under vegetated and partly vegetated patterns, the distributions of flow velocity and TKE are experimentally investigated in a laboratory open channel. Natural reed stems are used to simulate the rigid unsubmerged vegetation in the experiments. The following conclusions can be obtained:

1. The distributions of flow velocity and TKE can be significantly altered by the vegetation patterns in open channel. For the vegetated banks of vegetated and partly vegetated patterns, similar variation trends were observed for flow velocity although the variation for partly vegetated pattern was found gentler than that of vegetated pattern. Flow velocities are reduced before vegetation section where siltation may be induced, and then increased until the maximum value at the exit of vegetation section, where erosion may be induced.
2. It can be found that TKE is relatively lower and stable in the before and fore part of vegetation block. In the rear part of vegetation section, TKE is forced to increase rapidly and reach their maximum values near the exit of vegetation block. At the maximum location for TKE, a lot of closed concentric circles named recirculation cells are observed in the contour map. In addition, the maximum TKE for partly vegetated pattern which may induce larger potential soil erosion is moved forward 1 m compared with the vegetated pattern.
3. An empirical function $TKE = TKE_0 + a * \exp(-(y - y_c)^2 / 2b^2)$ is found to describe the variations of the average TKE along the flow direction, with which the TKE values may be predicted. The coefficients of determination are calculated as 0.995 ($p = 0.0001$) and 0.691 ($p = 0.0127$) corresponding to the vegetated banks under vegetated and partly vegetated patterns.
4. Vegetation pattern should be considered an important factor in the restoration of ecological waterways. In the present research, partly vegetated pattern is found better due to the gentler distributed flow velocity and lower TKE at the rear part of vegetation section.

In actual situations, more interactive factors affecting the effects of vegetation patterns on water flow should be considered, and it will be interesting and necessary to further consider the vegetation patterns and hydraulic conditions together. With the findings of the research, the comprehensive understanding on the effects of vegetation patterns on turbulence and erosion may be improved, and some inspirations to the practices of river ecological restoration may be provided.

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